



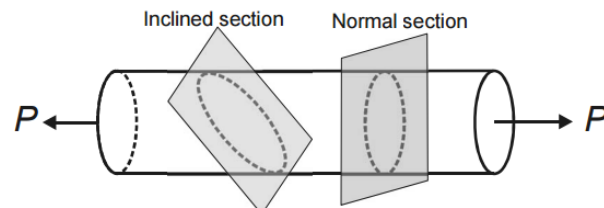
MENG 3050 - Power Transmission Components

Lesson 01-C-Stress and Deformation (review)

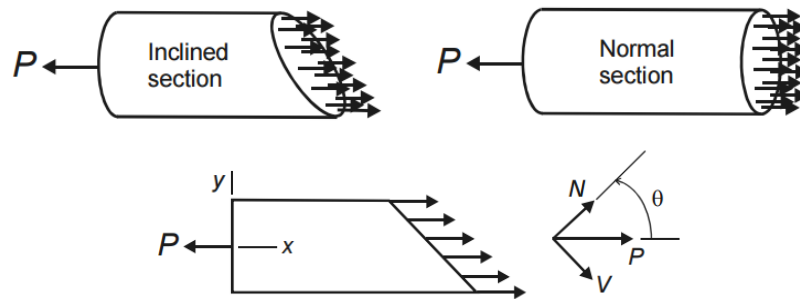
Introduction to stress elements:

Stress Transformation Equations:

To obtain a complete picture of the stresses in a bar, we must consider the stresses acting on an “inclined” (as opposed to a “normal”) section through the bar.



Because the stresses are the same throughout the entire bar, the stresses on the sections are uniformly distributed.

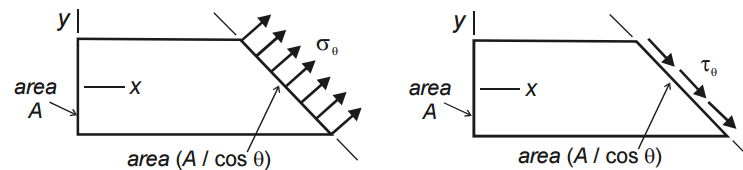


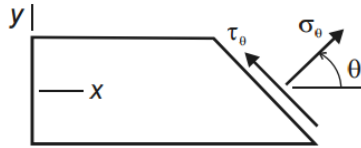
The force P can be resolved into components:

Normal force N perpendicular to the inclined plane, $N = P \cos \theta$

Shear force V tangential to the inclined plane $V = P \sin \theta$

If we know the areas on which the forces act, we can calculate the associated stresses





$$\sigma_{\theta} = \frac{\text{Force}}{\text{Area}} = \frac{N}{\text{Area}} = \frac{P \cos \theta}{A / \cos \theta} = \frac{P}{A} \cos^2 \theta$$

$$\sigma_{\theta} = \sigma_x \cos^2 \theta = \frac{\sigma_x}{2} (1 + \cos 2\theta)$$

$$\sigma_{\max} = \sigma_x \text{ occurs when } \theta = 0^\circ$$

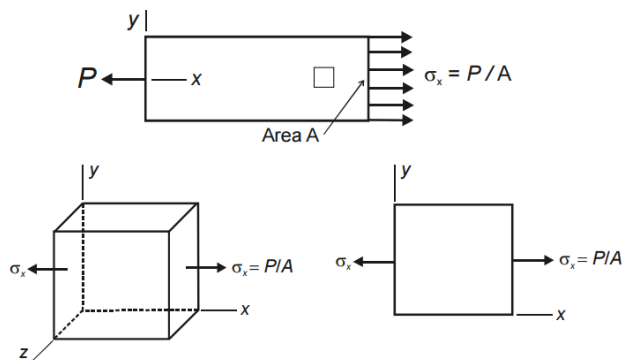
$$\tau_{\theta} = \frac{\text{Force}}{\text{Area}} = \frac{-V}{\text{Area}} = \frac{-P \sin \theta}{A / \cos \theta} = -\frac{P}{A} \sin \theta \cos \theta$$

$$\tau_{\theta} = -\sigma_x \sin \theta \cos \theta = -\frac{\sigma_x}{2} (\sin 2\theta)$$

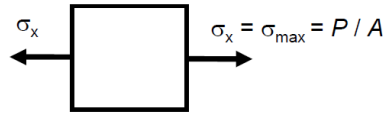
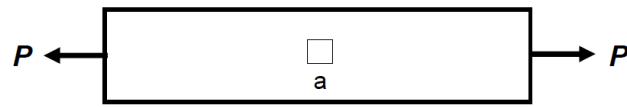
$$\tau_{\max} = \pm \sigma_x / 2 \text{ occurs when } \theta = \pm 45^\circ$$

Introduction to stress elements:

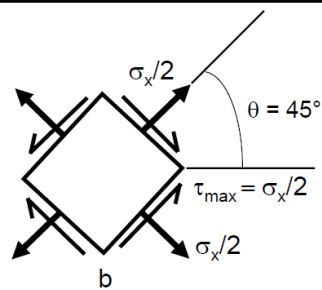
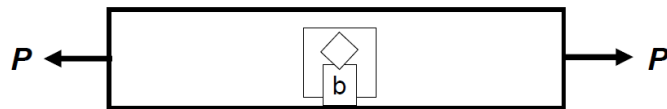
Stress elements are a useful way to represent stresses acting at some point on a body. Isolate a small element and show stresses acting on all faces. Dimensions are “infinitesimal”, but are drawn to a large scale.



Maximum stresses on a bar in tension:



**Maximum normal stress,
Zero shear stress**

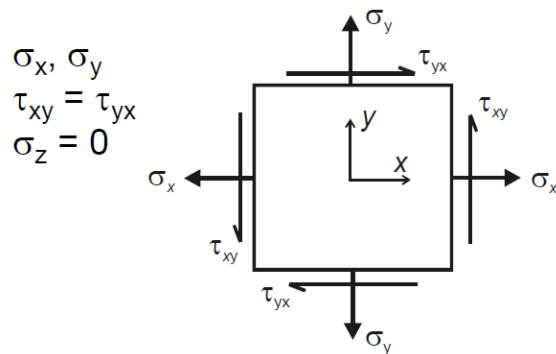


**Maximum shear stress,
Non-zero normal stress**

Plane stress element in 2D:

When an element is in plane stress in the xy plane, only the x and y faces are subjected to stresses ($\sigma_z = 0$ and $\tau_{zx} = \tau_{xz} = \tau_{zy} = \tau_{yz} = 0$).

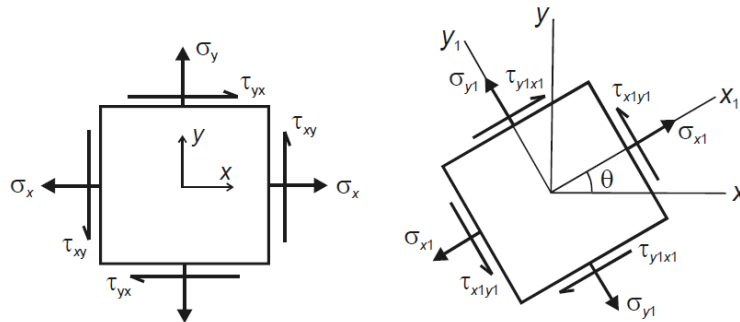
Such an element could be located on the free surface of a body (no stresses acting on the free surface).



Stresses on Inclined Sections:

A material may yield or fail at the maximum value of σ or τ . This value may occur at some angle other than $\theta = 0$.

(example for that that for uniaxial tension the maximum shear stress occurred when $\theta = 45$ degrees).



Transformation/ Equilibrium Equations:

Stresses	Forces
Forces can be found from stresses if the area on which the stresses act is known. Force components can then be summed	Left face has area A. Bottom face has area $A \tan \theta$. Inclined face has area $A \sec \theta$.

Simplify

$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

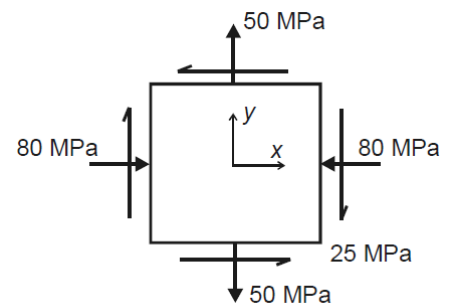
$$\tau_{x1y1} = -\frac{(\sigma_x - \sigma_y)}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Example: The state of plane stress at a point is represented by the stress element below. Determine the stresses acting on an element oriented 30° clockwise with respect to the original element.

Define the stresses in terms of the established sign convention:

$\sigma_x = -80$ MPa $\sigma_y = 50$ MPa. $\tau_{xy} = -25$ MPa, then find σ_{x1} , σ_{y1} , and τ_{x1y1} when $\theta = -30$

Solution: Substitute numerical values into the transformation equations:



$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

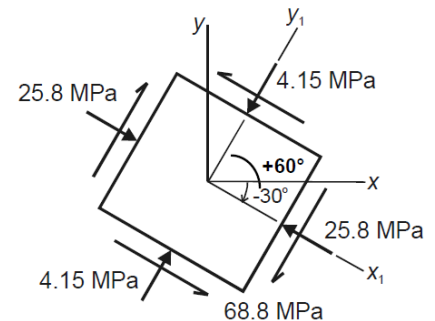
$$\sigma_{x1} = \frac{-80 + 50}{2} + \frac{-80 - 50}{2} \cos 2(-30^\circ) + (-25) \sin 2(-30^\circ) = -25.9 \text{ MPa}$$

$$\sigma_{y1} = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$

$$\sigma_{y1} = \frac{-80 + 50}{2} - \frac{-80 - 50}{2} \cos 2(-30^\circ) - (-25) \sin 2(-30^\circ) = -4.15 \text{ MPa}$$

$$\tau_{x1y1} = -\frac{(\sigma_x - \sigma_y)}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\tau_{x1y1} = -\frac{(-80 - 50)}{2} \sin 2(-30^\circ) + (-25) \cos 2(-30^\circ) = -68.8 \text{ MPa}$$



Principal Stresses

The maximum and minimum normal stresses (σ_1 and σ_2) are known as the principal stresses. To find the principal stresses, we must differentiate the transformation equations and equate to 0, results to:

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

θ_p are principal angles associated with the principal stresses
Principal stresses:

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2}$$

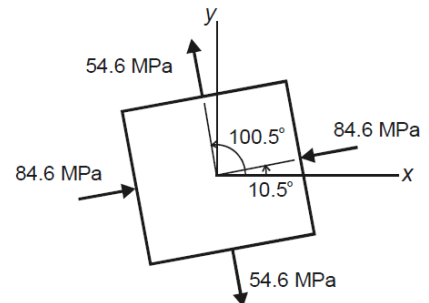
To verify which angle goes with which principal stress:

$$\sigma_{x1} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\sigma_{x1} = \frac{-80 + 50}{2} + \frac{-80 - 50}{2} \cos 2(10.5^\circ) + (-25) \sin 2(10.5^\circ) = -84.6 \text{ MPa}$$

$$\sigma_1 = 54.6 \text{ MPa with } \theta_{p1} = 100.5^\circ$$

$$\sigma_2 = -84.6 \text{ MPa with } \theta_{p2} = 10.5^\circ$$

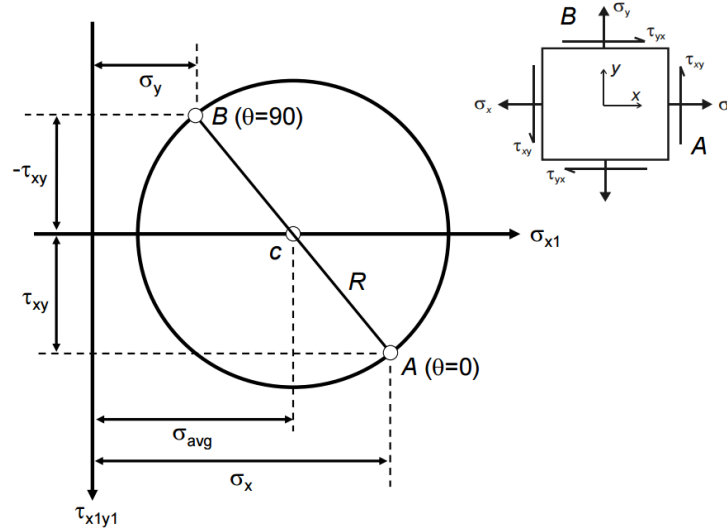


The two principal stresses determined so far are the principal stresses in the xy plane. However, the stress element is 3D, so there are always three principal stresses:

Mohr's Circle:

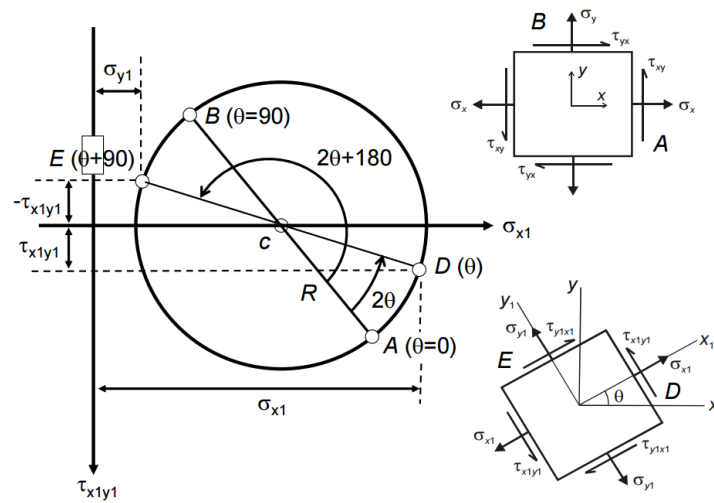
Procedure for Constructing Mohr's Circle:

1. Draw a set of coordinate axes with σ_{x1} as abscissa (positive to the right) and τ_{x1y1} as ordinate (positive downward).
2. Locate the centre of the circle c at the point having coordinates $\sigma_{x1} = \sigma_{avg}$ and $\tau_{x1y1} = 0$.
3. Locate point A, representing the stress conditions on the x face of the element by plotting its coordinates $\sigma_{x1} = \sigma_x$ and $\tau_{x1y1} = \tau_{xy}$. Note that point A on the circle corresponds to $\theta = 0^\circ$.
4. Locate point B, representing the stress conditions on the y face of the element by plotting its coordinates $\sigma_{x1} = \sigma_y$ and $\tau_{x1y1} = -\tau_{xy}$. Note that point B on the circle corresponds to $\theta = 90^\circ$.
5. Draw a line from point A to point B, a diameter of the circle passing through point c . Points A and B (representing stresses on planes at 90° to each other) are at opposite ends of the diameter (and therefore 180° apart on the circle).
6. Using point c as the centre, draw Mohr's circle through points A and B. This circle has radius R ,

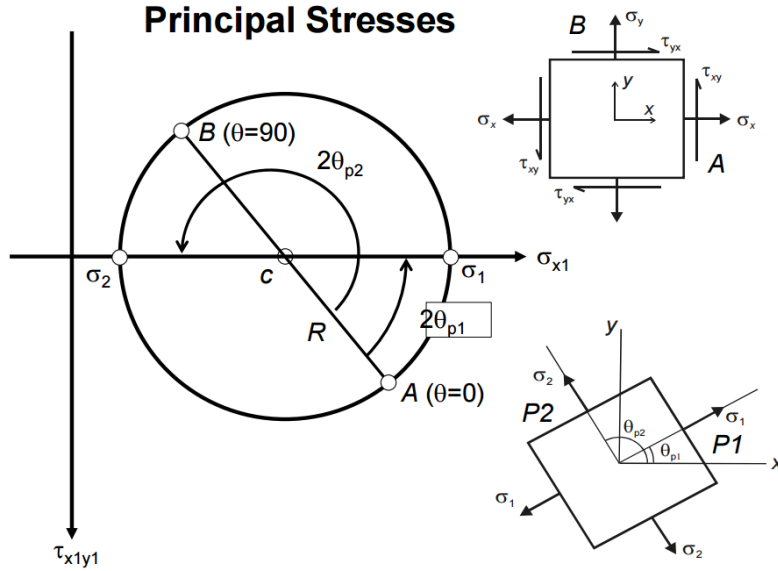


Stresses on an Inclined Element

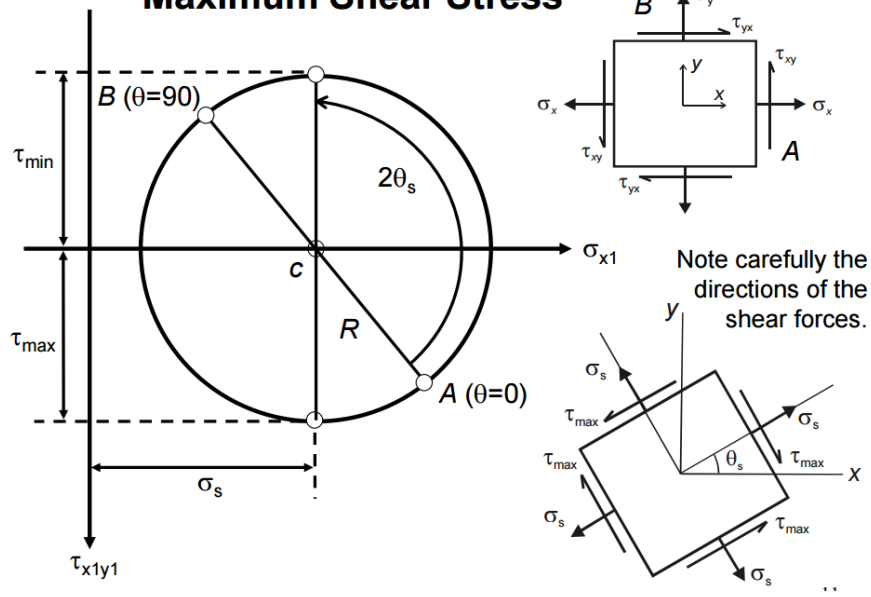
1. On Mohr's circle, measure an angle 2θ counter clockwise from radius cA , because point A corresponds to $\theta = 0$ and hence is the reference point from which angles are measured.
2. The angle 2θ locates the point D on the circle, which has coordinates σ_{x1} and τ_{x1y1} . Point D represents the stresses on the x_1 face of the inclined element.
3. Point E, which is diametrically opposite point D on the circle, is located at an angle $2\theta + 180^\circ$ from cA (and 180° from cD). Thus point E gives the stress on the y_1 face of the inclined element.
4. So, as we rotate the x_1y_1 axes counterclockwise by an angle θ , the point on Mohr's circle corresponding to the x_1 face moves counterclockwise through an angle 2θ .



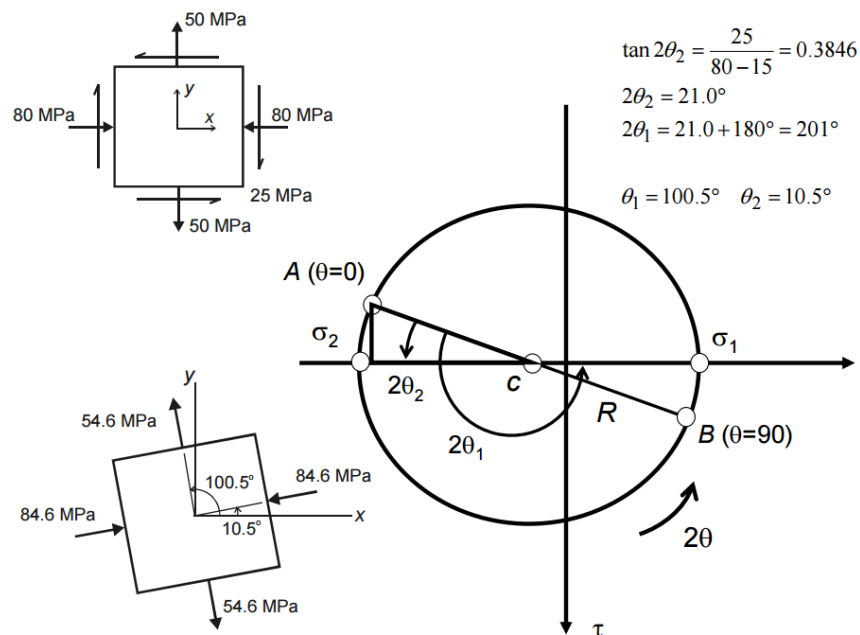
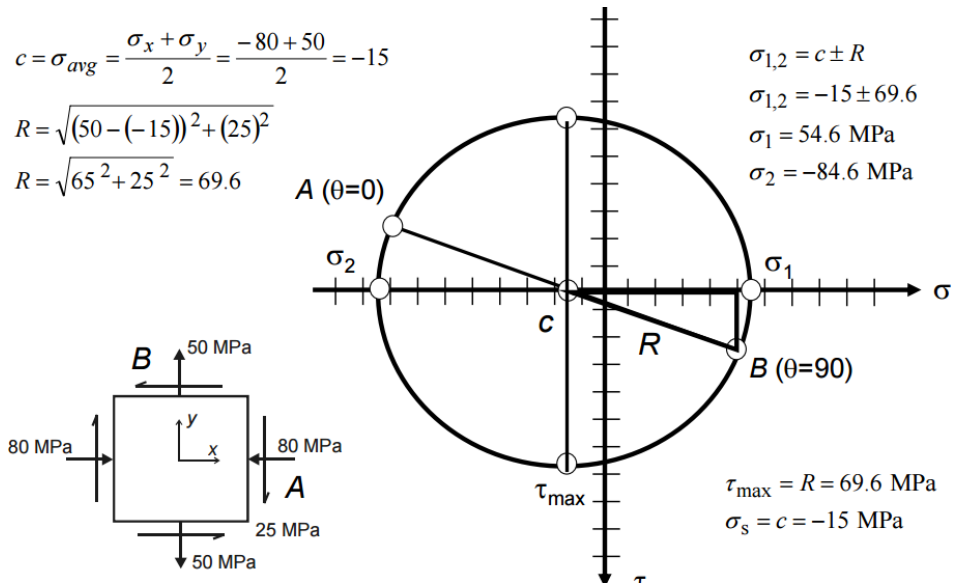
Principal Stresses

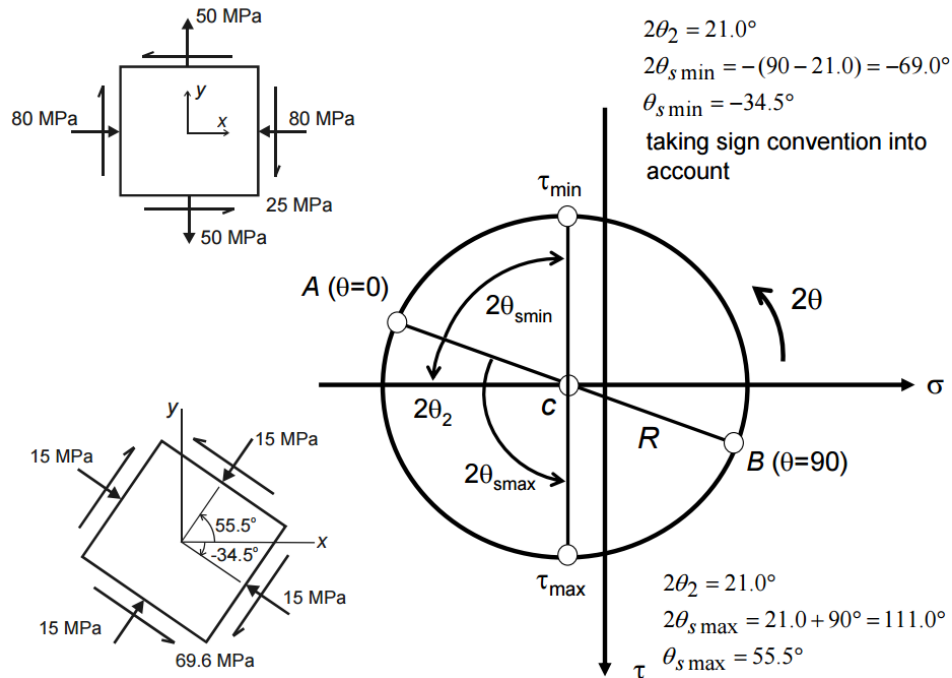


Maximum Shear Stress

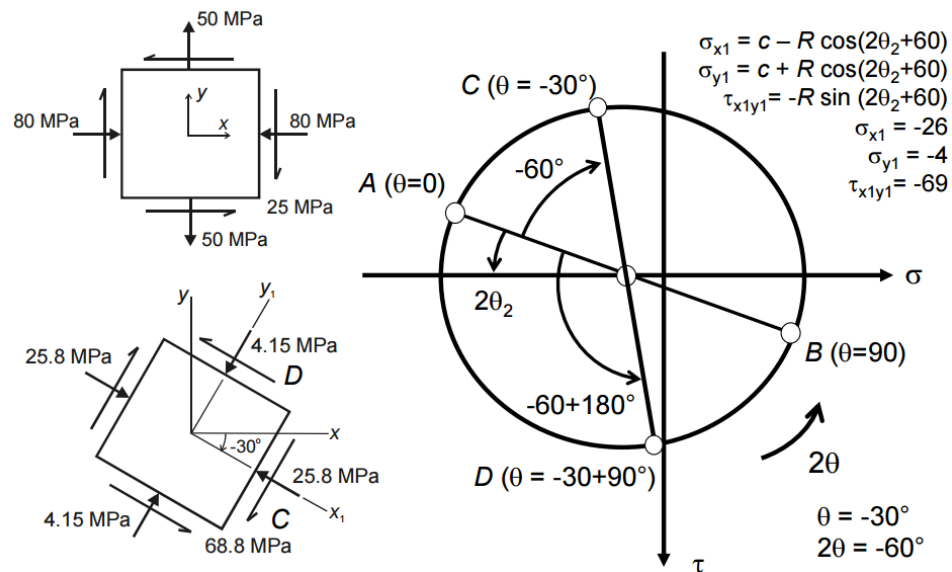


Example: The state of plane stress at a point is represented by the stress element below. Draw the Mohr's circle, determine the principal stresses and the maximum shear stresses, and draw the corresponding stress element



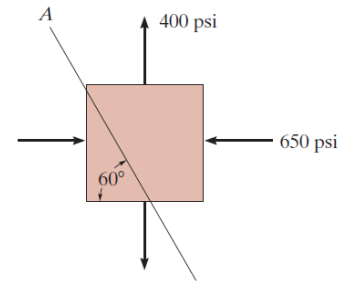


Example: The state of plane stress at a point is represented by the stress element below. Find the stresses on an element inclined at 30° clockwise and draw the corresponding stress elements.

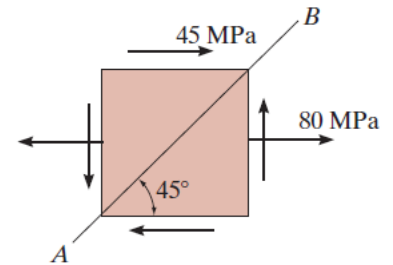


Assignments, Use Mohr's in all questions.

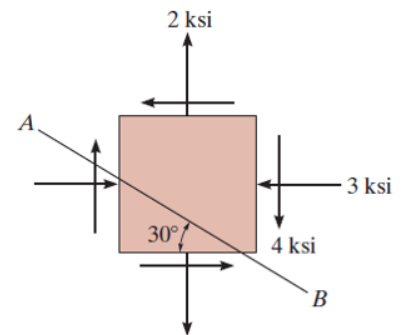
the states of stress at a point in a member is shown on the element. Determine the stress components acting on the inclined plane AB using the stress-transformation equations
solution:



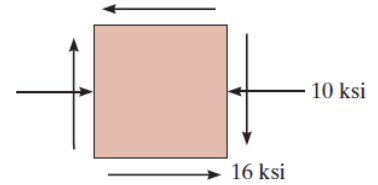
Determine the normal stress and shear stress acting on the inclined plane AB



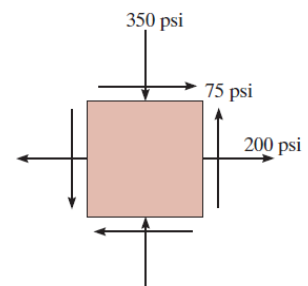
Determine the normal stress and shear stress acting on the inclined plane AB



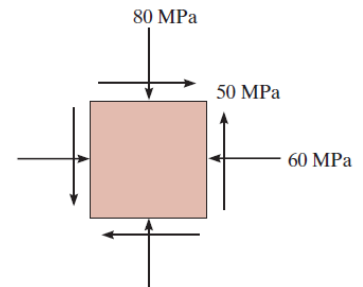
Determine the equivalent state of stress on an element if it is oriented 50° counterclockwise from the element shown.



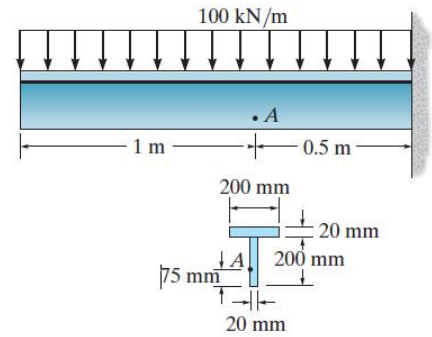
Determine the equivalent state of stress on an element if the element is oriented 60° clockwise from the element shown.



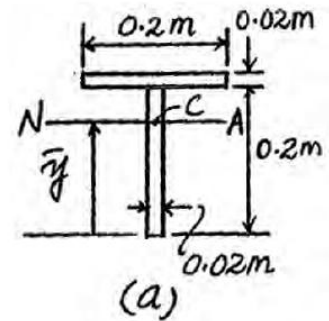
The state of stress at a point is shown on the element. Determine (a) the principal stress and (b) the maximum in-plane shear stress and. Specify the orientation of the element in each case. Show the results on each element.



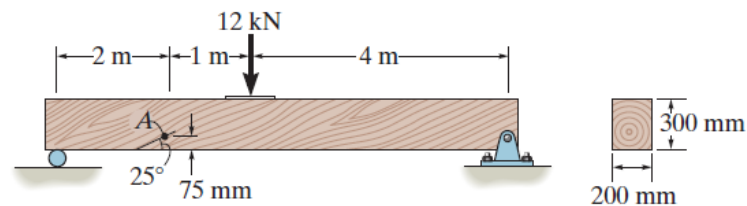
The T-beam is subjected to the distributed loading that is applied along its centerline. Determine the principal stress at point A and show the results on an element located at this point



The state of principal stress can be represented by the element shown in Fig. e.

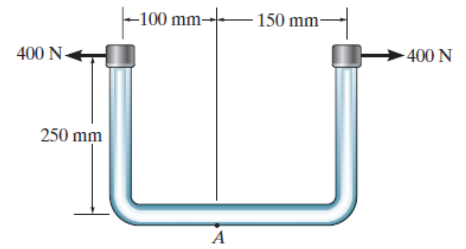


The wood beam is subjected to a load of 12 kN. If a grain of wood in the beam at point A makes an angle of 25° with the horizontal as shown, determine the normal and shear stress that act perpendicular and parallel to the grain due to the loading. Determine the principal stress at point A and specify the orientation of the element



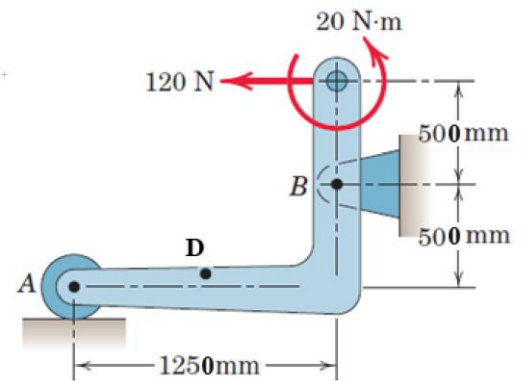
The bent rod has a diameter of 20 mm and is subjected to the force of 400 N. Determine the principal stress and the maximum in-plane shear stress that is developed at point A. Show the results on a properly

oriented element located at this point.

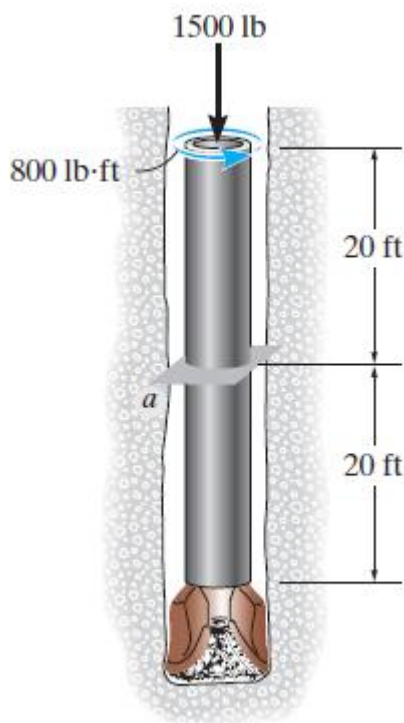


for the bell crank loaded and supported as shown. Calculate the magnitude of the force supported by the pin at B
Using Mohr circle, calculate the principle stresses at point D

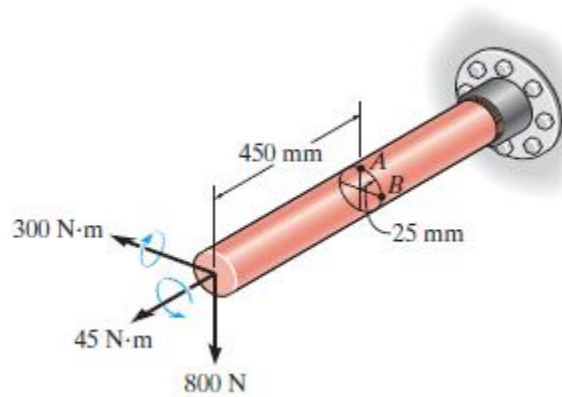
Assume that bell crank has a uniform cross section of 80mm x 80mm



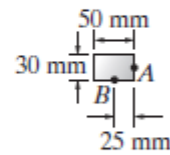
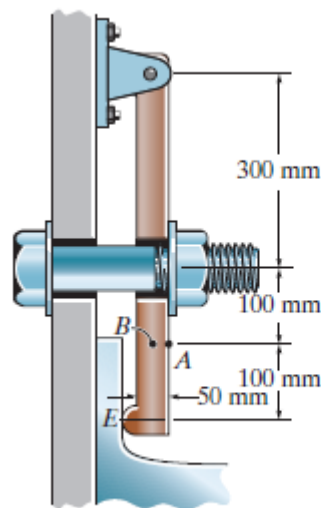
The drill pipe has an outer diameter of 3 in., a wall thickness of 0.25 in., and a weight of 50 lb/ft. If it is subjected to a torque and axial load as shown, determine (a) the principal stress and (b) the maximum in-plane shear stress at a point on its surface at section a.



The solid shaft is subjected to a torque, bending moment, and shear force as shown. Determine the principal stresses acting at point A.



The clamp bears down on the smooth surface at E by tightening the bolt. If the tensile force in the bolt is 40 kN, determine the principal stress at points A and B and show the results on elements located at each of these points. The cross-sectional area at A and B is shown in the adjacent figure



The 2-in.-diameter drive shaft AB on the helicopter is subjected to an axial tension of 10 000 lb and a torque of 300 lb.ft. Determine the principal stress and the maximum in-plane shear stress that act at a point on the surface of the shaft.

